Cleveland Clinic

Intrauterine Growth Restriction

Intrauterine growth restriction (IUGR) is when the fetal weight is estimated to be below the 10th percentile for its gestational age. Causes range from chronic diseases or infections in the mother to issues with the placenta and umbilical cord. Treatment includes frequent fetal monitoring and testing and, possibly, early delivery.

Overview

What is intrauterine growth restriction?

Intrauterine growth restriction (IUGR), also called fetal growth restriction (FGR), is when the fetus is smaller than it should be for its gestational age (number of weeks in the pregnancy). It means the fetus isn't growing at the expected rate while inside your uterus. IUGR is defined as weighing less than 9 out of 10 babies (below the 10th percentile) of the same age.

Fetuses grow at different rates, and "measuring small" doesn't necessarily mean the fetus has IUGR. In most cases, fetuses that measure small are born small, but healthy. It's similar to adults in that being 5'1" doesn't make you less healthy than someone who's 6'4". However, some mothers have an underlying medical condition that causes the fetus to measure small for its gestational age.

Types of intrauterine growth restriction

- Symmetric (or primary): All the fetal body parts are small in size. This accounts for up to 30% of IUGR cases.
- Asymmetric (or secondary): Just the fetus's abdomen measures small. The head and brain are the expected sizes. This accounts for up to 80% of all IUGR cases.

How common is fetal growth restriction?

Intrauterine growth restriction affects about 10% of all pregnancies. It's important to note that the fetus can be smaller than average and not have IUGR.

When is IUGR detected?

IUGR is often detected after 20 weeks of pregnancy when your healthcare provider begins measuring your fundal height (distance between your pubic bone and uterus) at each prenatal visit. If this measurement is shorter than anticipated, your provider may suspect IUGR and order more tests.

How does intrauterine growth restriction affect my pregnancy?

IUGR impacts your pregnancy in several ways. Your healthcare provider will want to keep track of fetal growth with additional ultrasounds, tests and prenatal appointments. If the cause of IUGR is related to a condition, your healthcare provider will help you manage the underlying condition. How your IUGR diagnosis is managed is unique to your pregnancy and condition.

Is IUGR a high-risk pregnancy?

Your healthcare provider may identify your pregnancy as high-risk depending on the cause of the growth restriction. However, most fetuses diagnosed with IUGR are healthy and simply require more monitoring during pregnancy.

Symptoms and Causes

What causes restricted fetal growth?

There are several possible causes of fetal growth restriction. In some cases, it develops because there's an issue with the placenta or umbilical cord. These organs are responsible for bringing nutrients, oxygen and blood to the fetus. Without the proper amount of nourishment, the fetus can be small for its age.

A fetus can develop FGR if you:

- Are expecting twins or triplets.
- Have had a previous baby with IUGR.
- Have high blood pressure, heart disease or low blood count.
- Smoke, drink alcohol or use recreational drugs.
- Have an autoimmune condition.
- Have kidney disease.
- Take anti-seizure medications.
- Have diabetes.
- Have an infection such as rubella, syphilis, cytomegalovirus (CMV) or toxoplasmosis.

Fetal growth restriction can also occur if your baby has a congenital or genetic disorder such as Down syndrome (trisomy 21). Living in high altitudes has also been associated with lower birth weight.

What is the most common cause of fetal growth restriction?

Fetal growth restriction can be caused by many factors or have no cause. Your healthcare provider will work with you to identify the underlying cause (if there is one).

What are the symptoms of intrauterine growth restriction?

Most people don't have any symptoms or signs that the fetus is small for its gestational age. You may feel that your belly or the fetus isn't as big as it should be. Even if you feel this way, only your healthcare provider can diagnose intrauterine growth restriction.

Diagnosis and Tests

How is intrauterine growth restriction diagnosed?

Intrauterine growth restriction is based on having an accurate gestational age or determining your due date. This is found by identifying the first day of your last menstrual period and by performing a first-trimester ultrasound. Having this date gives your healthcare provider a baseline for diagnosing IUGR.

Once your provider determines gestational age (or weeks in pregnancy), they can can diagnose IUGR using a few different methods.

Uterine fundal height

Fundal height is the size of your belly from the top of your pubic bone to the top of your uterus or the size of your "baby bump." Your healthcare provider measures your fundal height using measuring tape (usually made of paper). They place one end on your pubic bone and the other end at the top of your uterus.

Fundal height is measured in centimeters (cm) and should be about the same number you are in weeks of pregnancy (after 20 weeks). For example, if you are 32 weeks pregnant, your fundal height should be around 32 cm. Your healthcare provider may suspect IUGR if you measure at least 4 cm less than what your fundal height should be. So, in this example, your provider may suspect IUGR if your fundal height is 28 cm or lower.

Your provider can also use their hands to feel certain areas of your belly (called palpation) and estimate the size of the fetus.

Weight

Your healthcare provider measures your weight at every prenatal appointment. Poor weight gain could indicate that the fetus is also not gaining enough weight.

Ultrasound

If you are diagnosed with IUGR, you will have additional ultrasounds to monitor fetal growth throughout your pregnancy.

During a fetal ultrasound, your healthcare provider moves a wand or probe across your belly. Sound waves from the ultrasound create a picture of the fetus. Your healthcare provider uses the picture to take measurements and estimate the size and weight of the fetus. They can also use ultrasound to check the blood flow from the placenta through the umbilical cord or through the blood vessels (this is called Doppler flow). Poor circulation of blood can suggest IUGR.

Ultrasound can also measure the amount of amniotic fluid surrounding the fetus. Too little amniotic fluid could indicate IUGR.

Fetal monitoring

Your healthcare provider straps a monitoring device across your uterus. You lie down wearing this device for about 30 minutes while it tracks the fetal heart rate. Your healthcare provider checks these results for any signs of growth issues.

Amniocentesis

Amniocentesis can help determine congenital causes of IUGR. Your healthcare provider inserts a needle through your skin into your uterus. A sample of amniotic fluid is drawn into the needle and then tested for irregularities.

If your healthcare provider wants to run additional testing, try not to worry. Some fetuses are smaller than average but healthy and active.

Management and Treatment

How do you fix fetal growth restrictions?

Treatment for intrauterine growth restriction will depend on how far you are in your pregnancy and how healthy the fetus is. Unfortunately, there is not a "fix" for fetal growth restriction. Your healthcare provider will monitor the fetus closely to ensure there are no complications and treat you for any underlying conditions contributing to fetal growth restriction.

Treatments for fetal growth restriction

- **Increased monitoring:** Your healthcare provider will monitor you and the fetus more closely, checking on fetal growth with ultrasounds and other tests.
- **Early delivery:** If the pregnancy isn't developing or the fetus is at risk, labor may be induced early (before 37 weeks gestation). Sometimes a C-section is recommended because vaginal deliveries can stress growth-restricted babies.
- **Corticosteroid medication:** If your provider recommends an induction, they may give you medication to help your baby's lungs develop.

In rare or severe cases, your healthcare provider may admit you to the hospital for closer observation.

Do growth-restricted babies catch up?

Yes, most babies with IUGR will reach the size of full-term children their age by age 3. It's important to note that all children grow differently and that you shouldn't attempt to overfeed your child. This could unintentionally cause other conditions like obesity or diabetes. Your baby's pediatrician will work with you if they feel your baby's growth hasn't progressed as expected.

How can I increase fetal growth?

There isn't a guarantee that eating certain foods will increase fetal weight, but eating a healthy diet during pregnancy is certainly recommended. Eating fruits, vegetables, healthy carbohydrates and protein provides all the nutrients you and the fetus need during pregnancy. Talk to your healthcare provider about any significant changes to your diet during pregnancy.

Outlook / Prognosis

Does IUGR go away?

There's a chance the fetus will catch up a little. However, the diagnosis won't go away. Your healthcare provider will continue to closely monitor fetal growth for the remainder of your pregnancy.

Does fetal growth restriction cause early labor?

FGR doesn't usually cause early labor, but it's possible. Your healthcare provider may recommend inducing labor early if the fetus has stopped growing or if there are issues with your placenta or umbilical cord. They may also recommend a C-section if the stress of a vaginal delivery seems risky for your baby.

What are the risks to a baby born with IUGR?

Certain conditions and issues are associated with IUGR:

- Increased risk for C-section delivery and premature birth.
- Breathing and eating problems at birth.
- Lack of oxygen at birth (hypoxia).
- Low blood sugar at birth (hypoglycemia).

- Trouble regulating body temperature.
- Meconium aspiration (your baby swallows their first poop).
- Polycythemia (increased number of red blood cells).
- Problems fighting infection.

Some babies born early or small may need extra time at the hospital or special care in the neonatal intensive care unit (NICU).

There can be long-term effects of IUGR, depending on:

- The underlying condition that caused it.
- How restricted your baby's growth was.
- How many weeks gestation they were at delivery.

Your baby may be at increased risk for cognitive or developmental problems such as hyperactivity, cerebral palsy and poor test scores in school. They may also be at risk for obesity and cardiovascular disease later in life.

Prevention

How can I prevent intrauterine growth restriction?

Certain modifiable factors can put you at a higher risk for IUGR including:

- Using drugs and alcohol or smoking cigarettes.
- Eating a poor diet.
- Not gaining enough weight during pregnancy.

What health conditions put me at a higher risk for fetal growth restriction?

You may be at a higher risk for FGR if you have:

- High blood pressure or heart disease.
- Diabetes.
- Anemia or sickle cell anemia.
- Lupus or other autoimmune conditions.

Living With

What do I do if my baby has IUGR?

First, don't panic. You likely didn't do anything to cause IUGR. There are a few things you can do to help:

- Make sure to keep all your prenatal appointments and fetal testing appointments (like ultrasounds).
- If you're using drugs, smoking or drinking alcohol, stop right away.
- Eat a healthy diet and enough calories each day.
- Get enough rest and sleep eight hours per night.
- Pay attention to decreases in fetal movement.
- Be honest with your healthcare provider about any medications you're taking.

Work with your healthcare provider to make sure you understand the diagnosis, potential complications and how you can modify your lifestyle.

Should I worry if my baby is measuring small?

Most of the time, no. Fetuses can grow at different rates. However, there's a difference between the fetus measuring small and a diagnosis of fetal growth restriction. FGR is diagnosed when the fetus weighs less than 9 out of 10 fetuses of the same age. It's possible for the fetus to measure small but still be healthy and not at an increased risk for complications. Talk to your healthcare provider about any concerns you have.

What questions should I ask my doctor?

It's normal to have questions about an intrauterine growth restriction diagnosis. Some questions to ask your healthcare provider are:

- Do you know what caused IUGR?
- How often do I need prenatal appointments?
- Can I do anything to help my baby grow?
- Is my baby at risk for any medical conditions?
- Does my baby need to stay in the hospital longer after birth?
- Will my baby have any health conditions later in life?

A note from Cleveland Clinic

A lot of factors can contribute to the fetus being small before birth. Your healthcare provider will determine if there's an underlying problem preventing the fetus from growing. In some cases, there may be no cause, and the fetus is born smaller than average. If the fetus is diagnosed with IUGR, your healthcare team will closely monitor fetal growth throughout the remainder of your pregnancy. Their priority is to ensure you and the fetus have the care and treatment you need. Most babies born with IUGR don't have health complications.

Experts You Can Trust

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References

Cleveland Clinic's health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

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